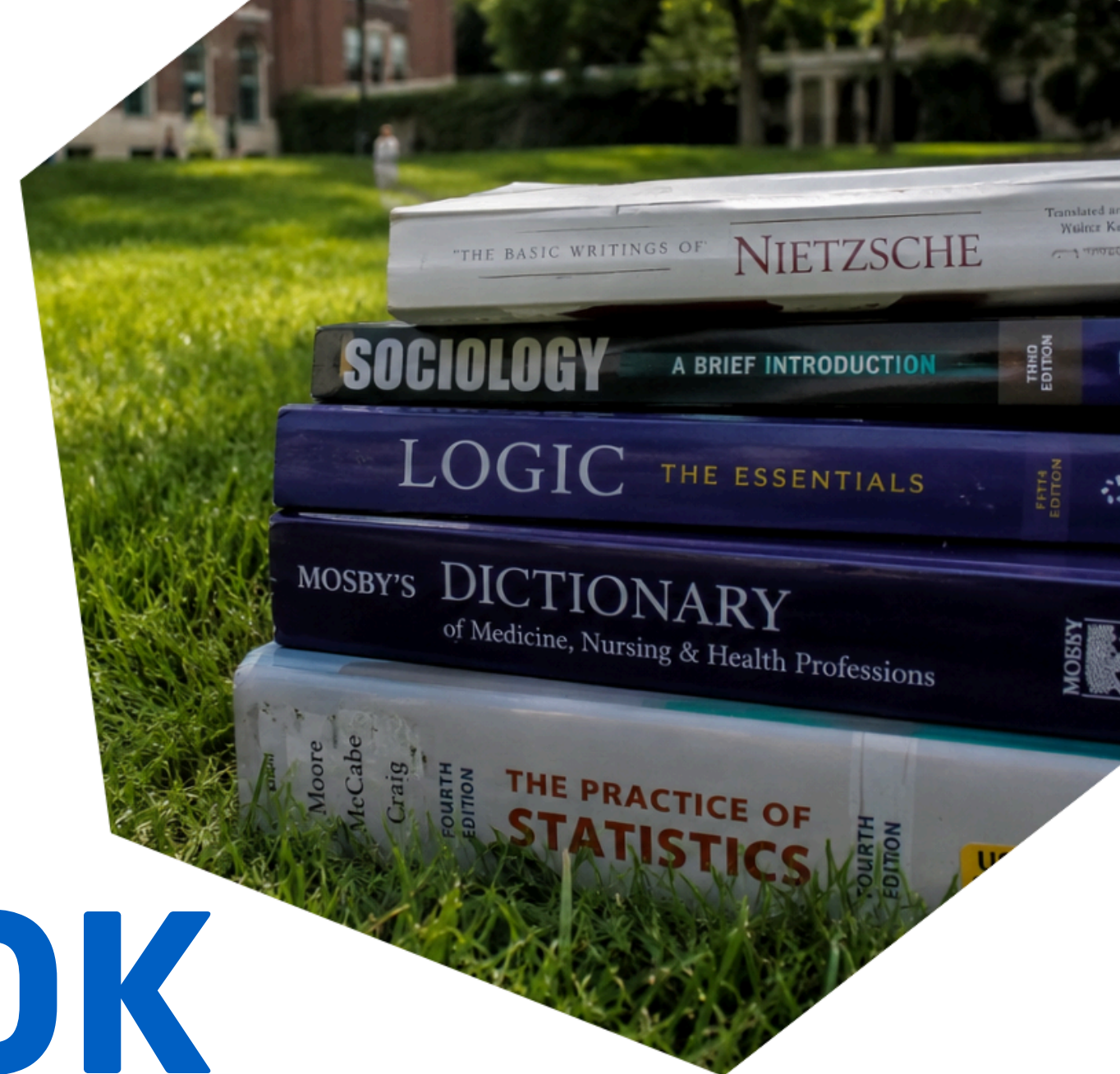
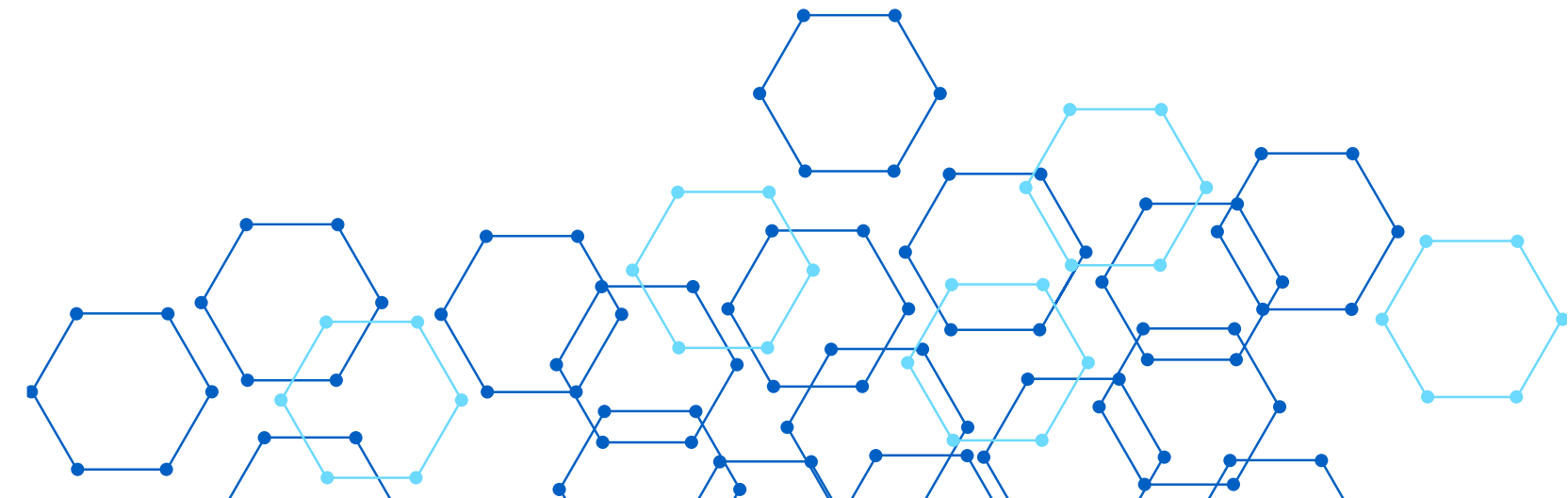


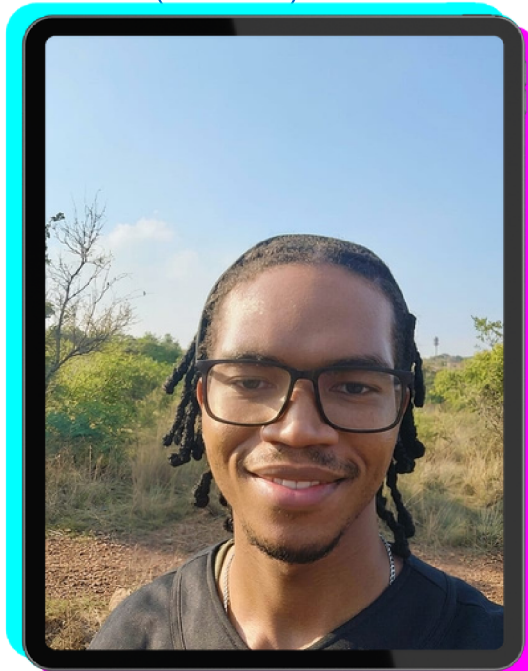
Presented by:



# UNI TEXTBOOK MARKETPLACE

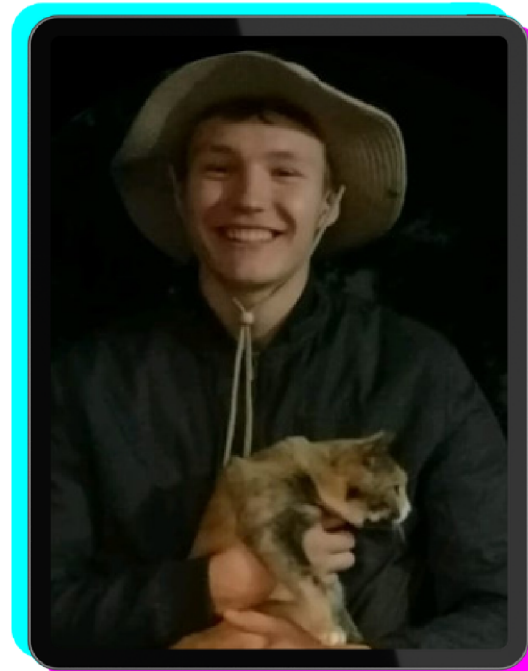


# MEET THE TEAM



**TIEGO MOKWENA**

Project Manager +  
DevOps + UI  
Engineer



**JOSH KRETSCMER**

Services Engineer  
2 + Integration  
Engineer 2



**GIFT MOHUBA**

Services Engineer  
1 + Integration  
Engineer 1



**NEO BOSOGA**

Data Engineer +  
Tester



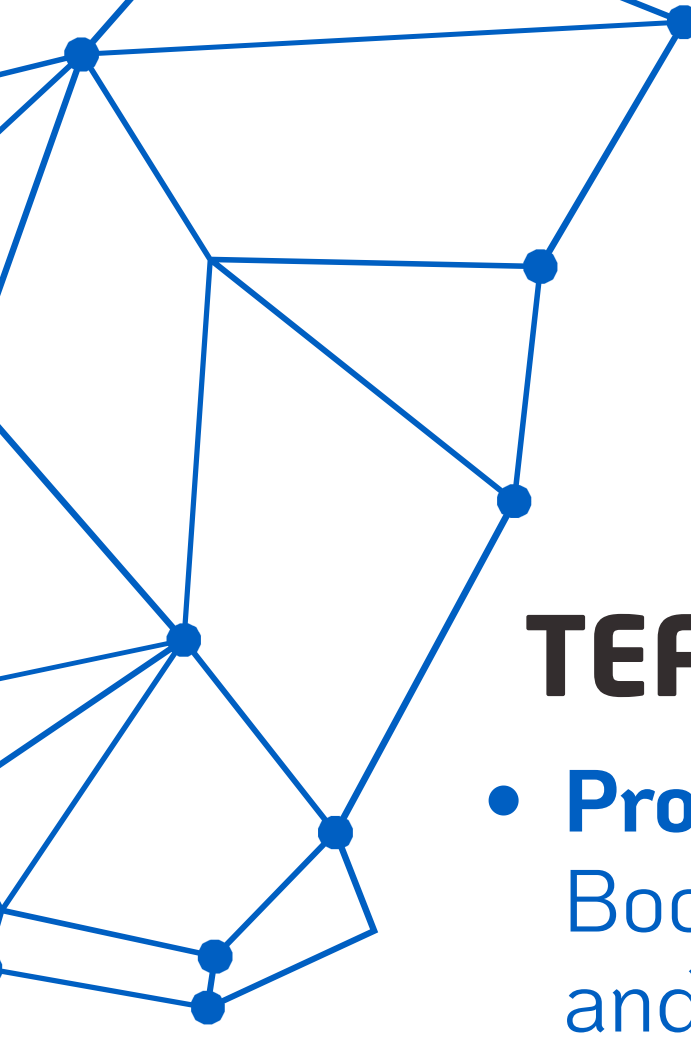
# A UNIFIED SPACE TO EASILY GET TEXTBOOKS



A university-verified marketplace with structured listings, module browsing, smart filters, and privacy-first messaging.

Students struggle to buy/sell textbooks on WhatsApp and Facebook. Listings lack details. Sellers are unverified. Scams are common.





# PROJECT PLAN

## AGILE SCRUM

### TEAM ROLES

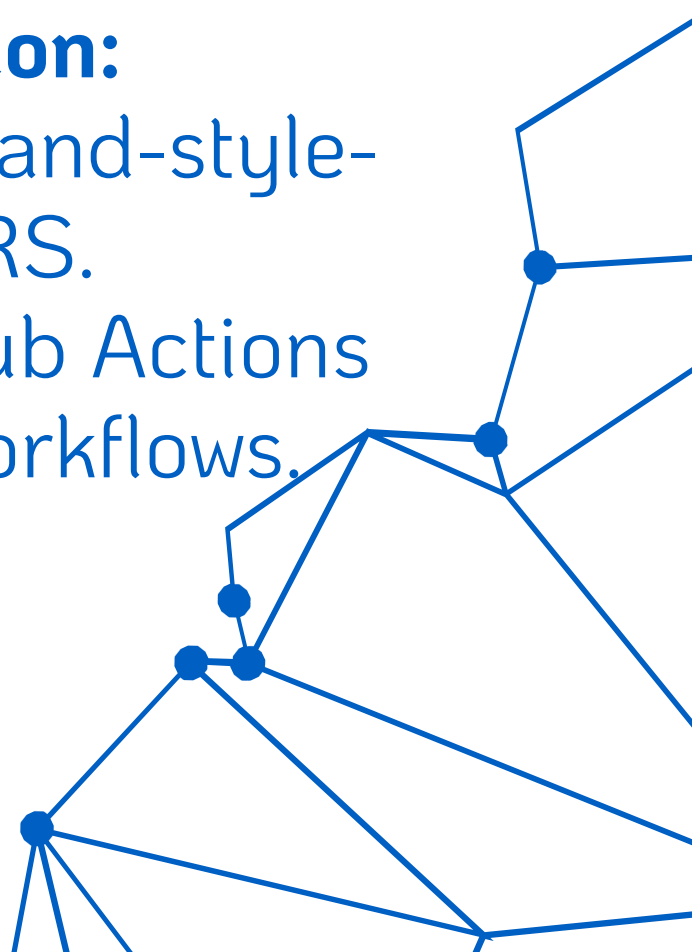
- **Product Owner:** Monet Booyesen, Kyle Marshall and Dominique da Silva
  - **Scrum Master:** Tiego Mokwena (Project Manager)
  - **Development Team:** NexusDev
- 



### SPRINT

- **Weekly Sprints**
- Sprint Planning
- Daily standups
- Sprint Review
- Sprint Retrospective

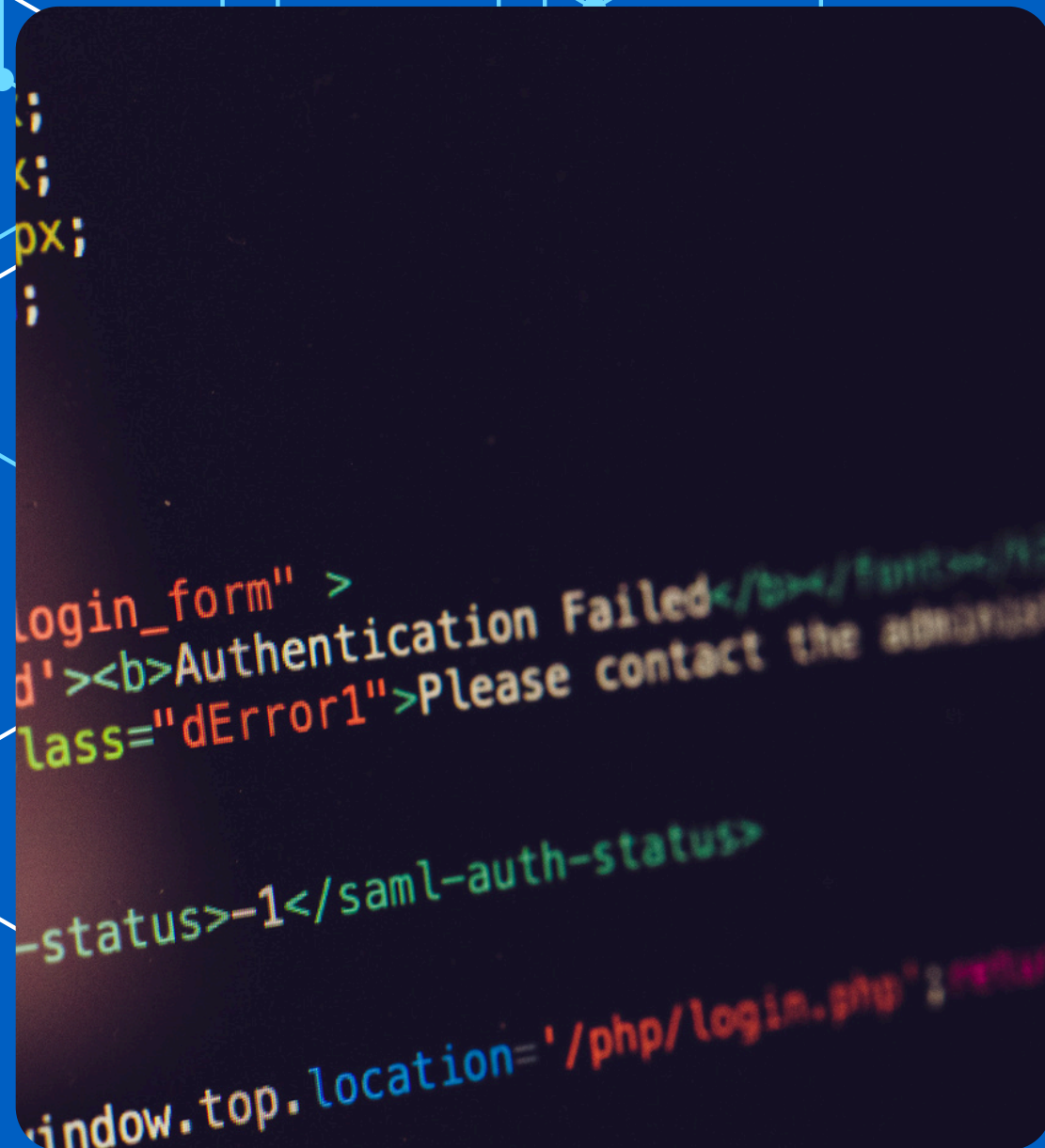
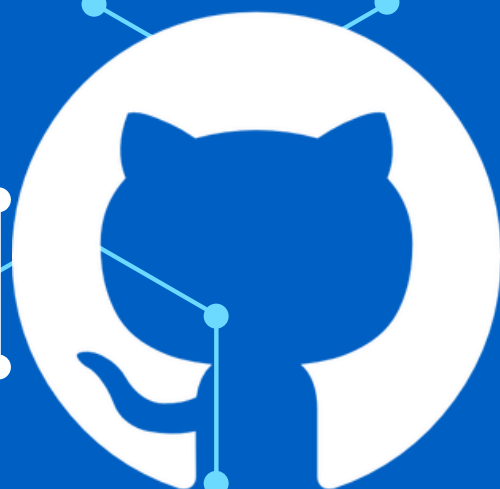
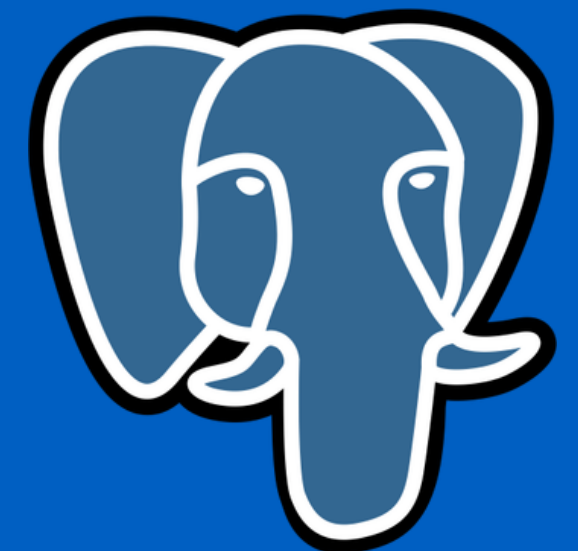
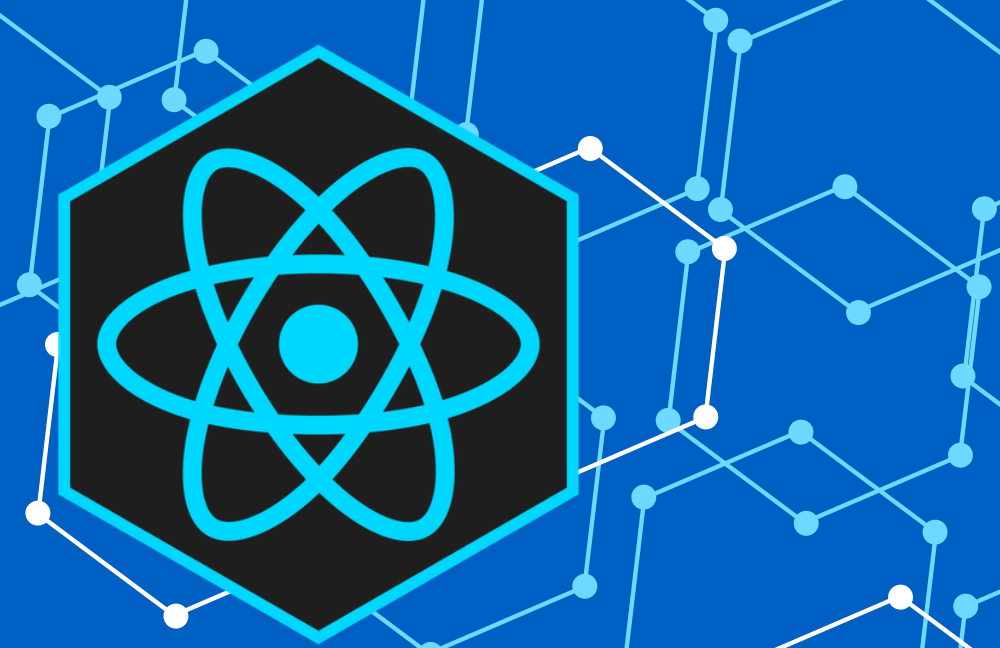
### TOOLS & PRACTICES

- **Project Tracking:** GitHub Projects/Issues
  - **Version Control:** Git(main, develop, feature)
  - **Documentation:** README, Brand-style-guide and SRS.
  - **CI/CD:** GitHub Actions for testing workflows.
- 
- 



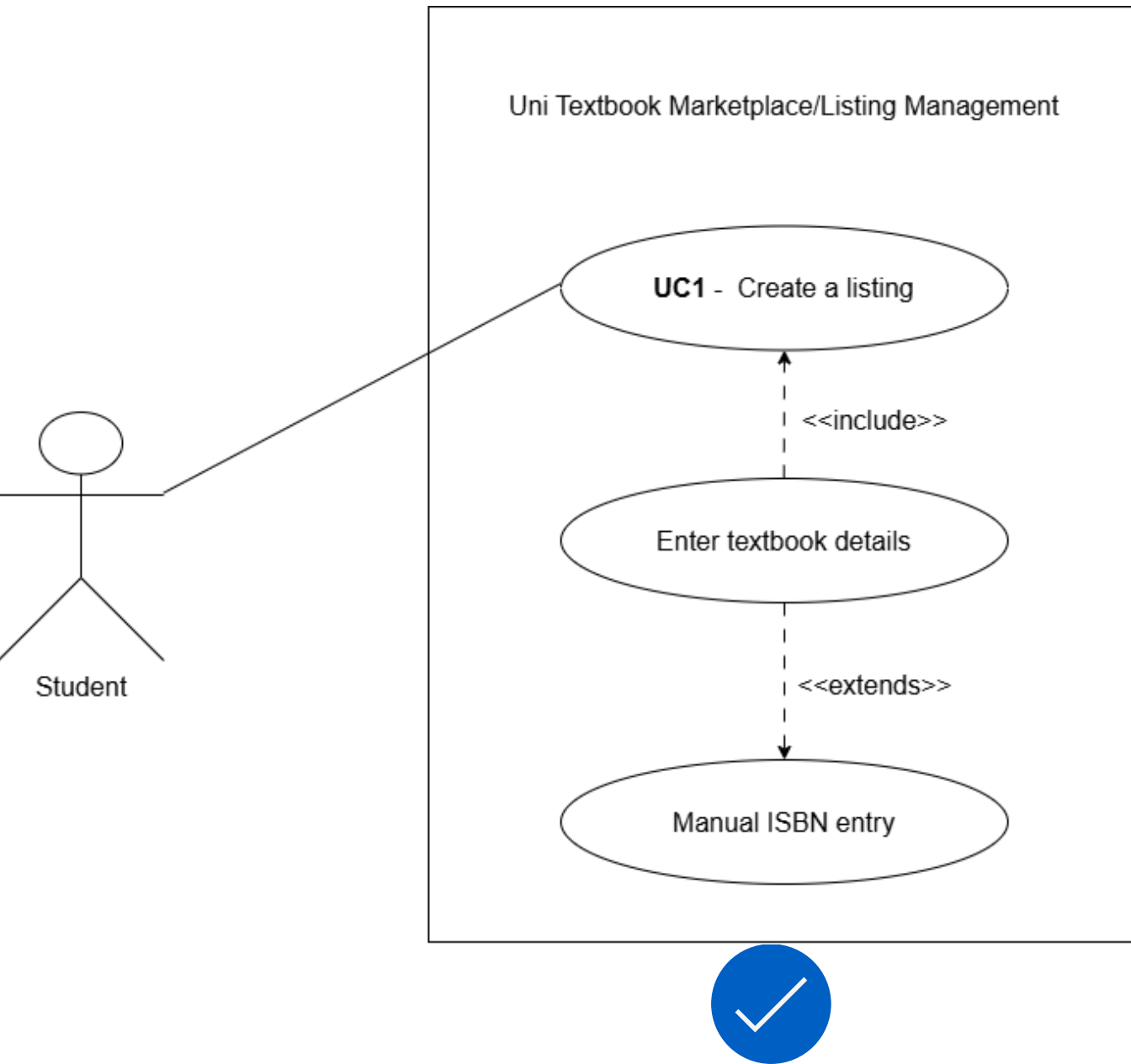
# SYSTEM ARCHITECTURE

| COMPONENT | TECHNOLOGY            |
|-----------|-----------------------|
| Frontend  | NextJS + Tailwind CSS |
| Backend   | NestJS                |
| Database  | PostgreSQL            |
| Auth      | JWT                   |
| CI/CD     | GitHub Actions        |

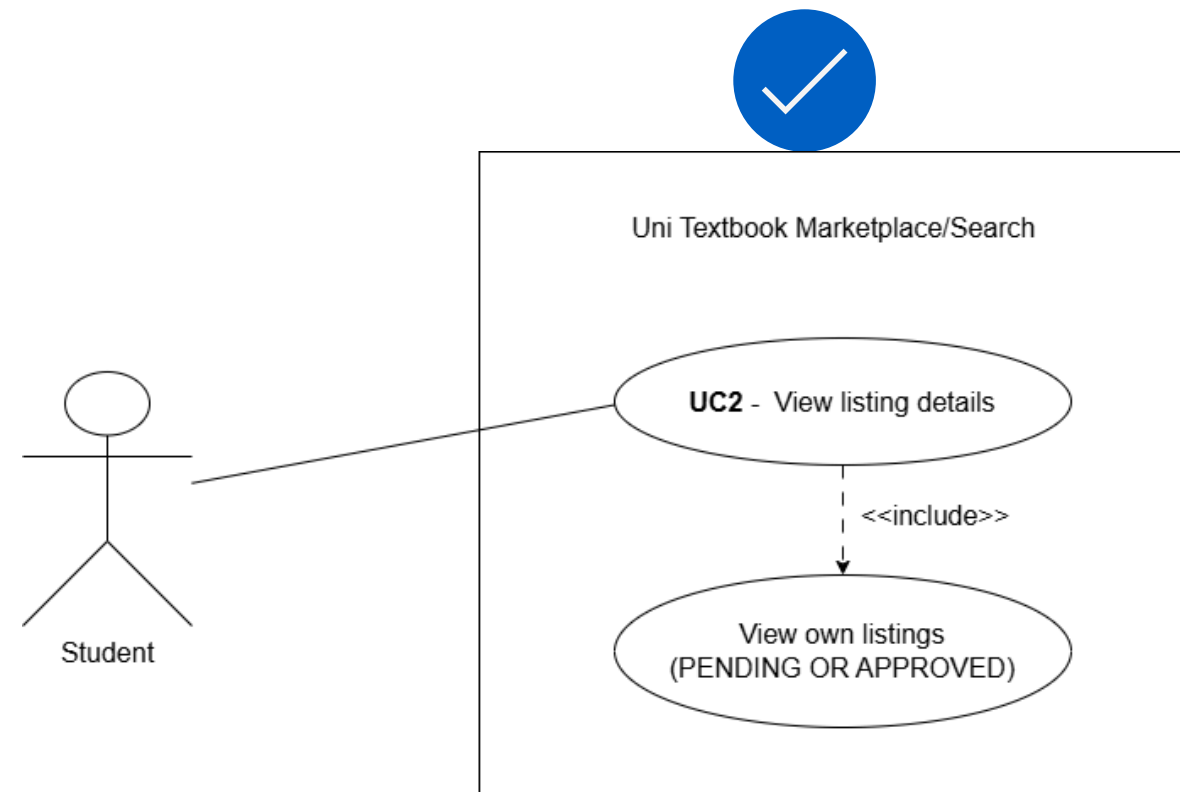


# USE CASES

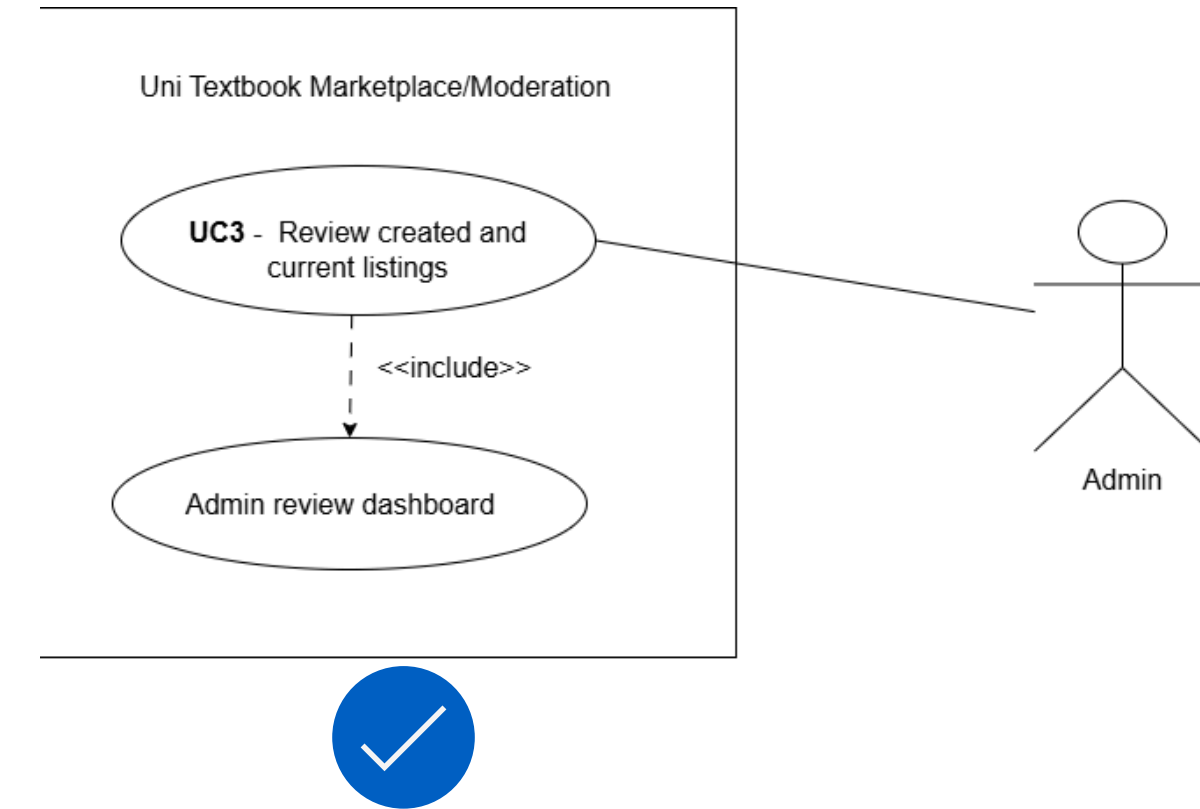
## UC1 CREATE A LISTING



## UC2 VIEW LISTING DETAILS

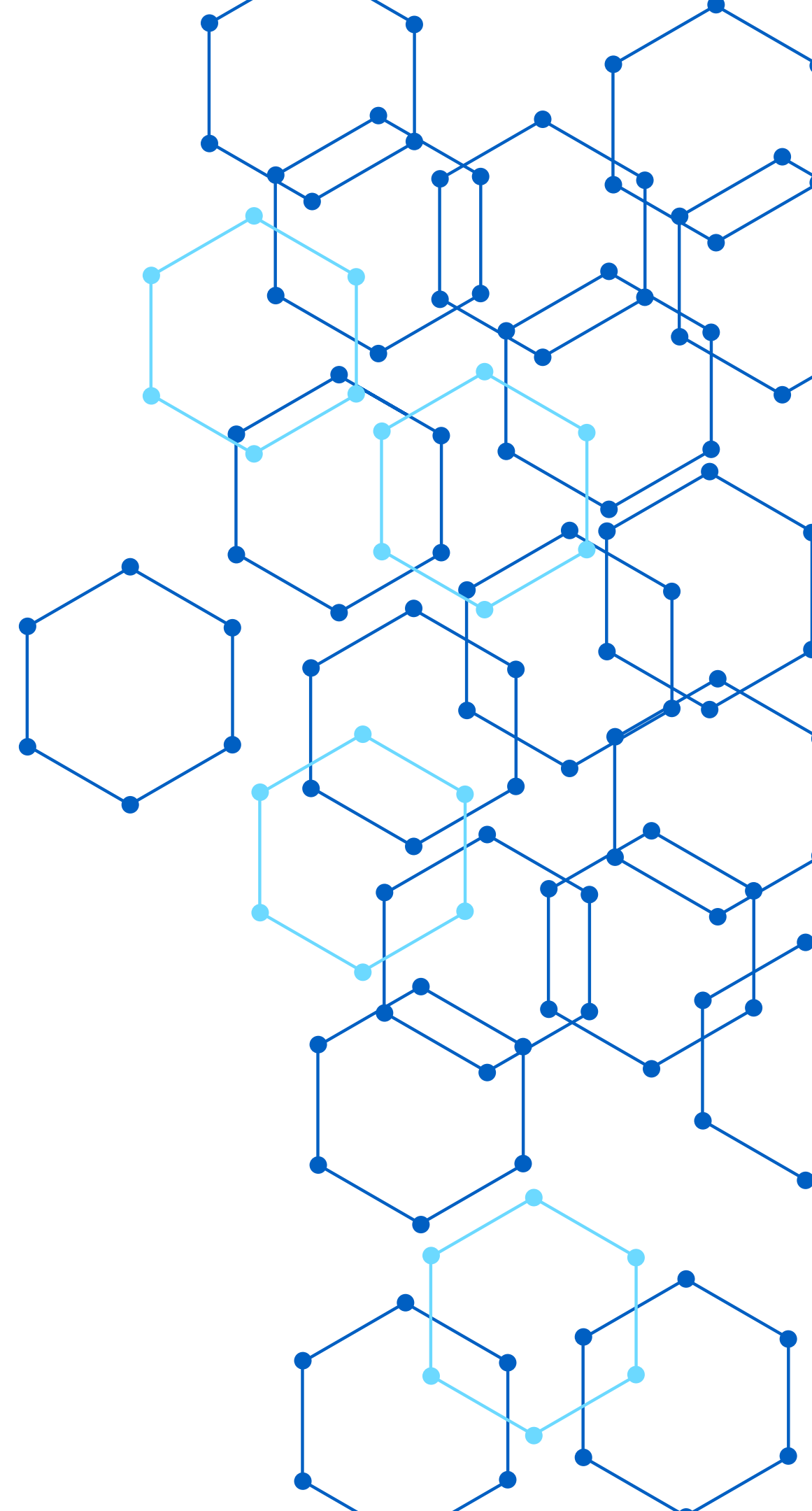


## UC3 REVIEW LISTINGS





# LIVE DEMO



# CI PIPELINE

```
jobs:
  # =====
  # Backend - NestJS
  # =====
  backend:
    name: Backend (NestJS)
    runs-on: ubuntu-latest

    services:
      postgres:
        image: postgres:15
        env: ...
        ports:
          - 5432:5432
        options: >-
          --health-cmd pg_isready
          --health-interval 10s
          --health-timeout 5s
          --health-retries 5

    steps:
      - name: Checkout code
        uses: actions/checkout@v4

      - name: Set up Node.js
        uses: actions/setup-node@v4
        with:
```

```
# =====
# Frontend - Next.js
# =====
frontend:
  name: Frontend (Next.js)
  runs-on: ubuntu-latest

  steps:
    - name: Checkout code
      uses: actions/checkout@v4

    - name: Set up Node.js
      uses: actions/setup-node@v4
      with:
        node-version: '20'

    - name: Remove lockfiles for clean Linux install
      run: |
        rm -f package-lock.json
        rm -f frontend/package-lock.json

    - name: Install root dependencies
      run: npm install

    - name: Lint frontend
      working-directory: frontend
```



# CI PIPELINE

## Actions

[New workflow](#)

All workflows

NexusDev CI

Management

 Caches

 Attestations 

 Runners

 Usage metrics 

 Performance metrics 

|   |                                                                                                                                                         |                          |                                 |     |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------------------------|-----|
| ✓ | <b>Merge pull request #53 from COS301-SE-2026/fe...</b><br>NexusDev CI #136: Commit <a href="#">36f5a64</a> pushed by <a href="#">tl21thebe</a>         | develop                  | May 20, 8:00 PM GMT+2<br>1m 38s | ... |
| ✓ | <b>feat: intergrated frontend and backend for login...</b><br>NexusDev CI #135: Pull request <a href="#">#53</a> synchronize by <a href="#">GiftMHB</a> | feature/AuthIntergration | May 20, 7:21 PM GMT+2<br>6m 52s | ... |
| ✗ | <b>feat: intergrated frontend and backend for login...</b><br>NexusDev CI #134: Pull request <a href="#">#53</a> synchronize by <a href="#">GiftMHB</a> | feature/AuthIntergration | May 20, 6:51 PM GMT+2<br>2m 9s  | ... |
| ✓ | <b>Merge pull request #48 from COS301-SE-2026/fe...</b><br>NexusDev CI #133: Commit <a href="#">da40e62</a> pushed by <a href="#">tl21thebe</a>         | develop                  | May 20, 6:47 PM GMT+2<br>3m 25s | ... |
| ✗ | <b>feat: intergrated frontend and backend for login...</b><br>NexusDev CI #132: Pull request <a href="#">#53</a> synchronize by <a href="#">GiftMHB</a> | feature/AuthIntergration | May 20, 6:46 PM GMT+2<br>4m 34s | ... |
| ✓ | <b>feat : seeding data</b><br>NexusDev CI #131: Pull request <a href="#">#48</a> synchronize by <a href="#">u23591732</a>                               | feature/schema           | May 20, 6:35 PM GMT+2<br>2m 57s | ... |
| ✓ | <b>Merge pull request #52 from COS301-SE-2026/fe...</b><br>NexusDev CI #130: Commit <a href="#">1668e32</a> pushed by <a href="#">tl21thebe</a>         | develop                  | May 20, 5:43 PM GMT+2<br>1m 27s | ... |

# TESTING & COVERAGE

**PASS** src/auth/auth.service.spec.ts (79.472 s)

A worker process has failed to exit gracefully and has been force exited. This is likely caused by tests leaking due to improper teardown. Try running with `--detectOpenHandles` to find leaks. Active timers can also cause this, ensure that `.unref()` was called on them.

| File               | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s |
|--------------------|---------|----------|---------|---------|-------------------|
| All files          | 82.11   | 79.41    | 75      | 82.67   |                   |
| auth               | 86.25   | 90       | 90.9    | 86.84   |                   |
| auth.controller.ts | 97.14   | 100      | 88.88   | 96.96   | 120               |
| auth.module.ts     | 0       | 100      | 0       | 0       | 1-40              |
| auth.service.ts    | 97.29   | 86.36    | 100     | 97.22   | 125,235           |
| otp.service.ts     | 100     | 100      | 100     | 100     |                   |
| auth/decorator     | 0       | 100      | 0       | 0       |                   |
| roles.decorator.ts | 0       | 100      | 0       | 0       | 1-3               |
| auth/dto           | 85.18   | 100      | 100     | 85.18   |                   |
| login.dto.ts       | 100     | 100      | 100     | 100     |                   |
| register.dto.ts    | 100     | 100      | 100     | 100     |                   |
| resend-otp.dto.ts  | 100     | 100      | 100     | 100     |                   |
| university.dto.ts  | 0       | 100      | 100     | 0       | 1-12              |
| verify-otp.dto.ts  | 100     | 100      | 100     | 100     |                   |
| auth/guards        | 64.7    | 0        | 50      | 53.84   |                   |
| jwt-auth.guard.ts  | 100     | 100      | 100     | 100     |                   |
| roles.guard.ts     | 50      | 0        | 50      | 40      | 20-30             |
| auth/strategies    | 70      | 0        | 0       | 62.5    |                   |
| jwt.strategy.ts    | 70      | 0        | 0       | 62.5    | 22-34             |

Test Suites: 2 passed, 2 total

Tests: 45 passed, 45 total

Snapshots: 0 total

Time: 130.336 s

Ran all test suites matching src/auth/.

# TESTING & COVERAGE

improper teardown. Try running with `--detectOpenHandles` to find leaks. Active timers can also cause this, ensure that `.unref()` was called on them.

| File                | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s |
|---------------------|---------|----------|---------|---------|-------------------|
| All files           | 84.88   | 85.41    | 80.55   | 85.23   |                   |
| auth                | 86.25   | 90       | 90.9    | 86.84   |                   |
| auth.controller.ts  | 97.14   | 100      | 88.88   | 96.96   | 120               |
| auth.module.ts      | 0       | 100      | 0       | 0       | 1-40              |
| auth.service.ts     | 97.29   | 86.36    | 100     | 97.22   | 125,235           |
| otp.service.ts      | 100     | 100      | 100     | 100     |                   |
| auth/decorator      | 0       | 100      | 0       | 0       |                   |
| roles.decorator.ts  | 0       | 100      | 0       | 0       | 1-3               |
| auth/dto            | 85.18   | 100      | 100     | 85.18   |                   |
| login.dto.ts        | 100     | 100      | 100     | 100     |                   |
| register.dto.ts     | 100     | 100      | 100     | 100     |                   |
| resend-otp.dto.ts   | 100     | 100      | 100     | 100     |                   |
| university.dto.ts   | 0       | 100      | 100     | 0       | 1-12              |
| verify-otp.dto.ts   | 100     | 100      | 100     | 100     |                   |
| auth/guards         | 64.7    | 0        | 50      | 53.84   |                   |
| jwt-auth.guard.ts   | 100     | 100      | 100     | 100     |                   |
| roles.guard.ts      | 50      | 0        | 50      | 40      | 20-30             |
| auth/strategies     | 70      | 0        | 0       | 62.5    |                   |
| jwt.strategy.ts     | 70      | 0        | 0       | 62.5    | 22-34             |
| listings            | 100     | 100      | 100     | 100     |                   |
| listings.service.ts | 100     | 100      | 100     | 100     |                   |

Test Suites: 3 passed, 3 total  
Tests: 66 passed, 66 total  
Snapshots: 0 total  
Time: 95.767 s

Ran all test suites matching `src/auth/|src/listings/listings.service.spec.ts`.



# TESTING & COVERAGE

```
getMyListings
  ✓ should return only listings belonging to the authenticated user (4 ms)
  ✓ should return empty array when user has no listings (4 ms)
  ✓ should not return listings from other users (2 ms)
getListingById
  ✓ should return listing when valid ID is provided (2 ms)
  ✓ should throw NotFoundException when listing does not exist (5 ms)
  ✓ should throw NotFoundException for empty string ID (3 ms)
getPendingListings (Admin only)
  ✓ should return all listings with PENDING status (2 ms)
  ✓ should return empty array when no pending listings exist (2 ms)
approveListing (Admin only)
  ✓ should approve a pending listing (8 ms)
  ✓ should throw NotFoundException when listing does not exist (3 ms)
rejectListing (Admin only)
  ✓ should reject a pending listing (2 ms)
  ✓ should throw NotFoundException when listing does not exist (6 ms)
```

| File                | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s |
|---------------------|---------|----------|---------|---------|-------------------|
| All files           | 100     | 100      | 100     | 100     |                   |
| listings.service.ts | 100     | 100      | 100     | 100     |                   |

```
Test Suites: 1 passed, 1 total
Tests:       21 passed, 21 total
Snapshots:   0 total
Time:        60.684 s
```

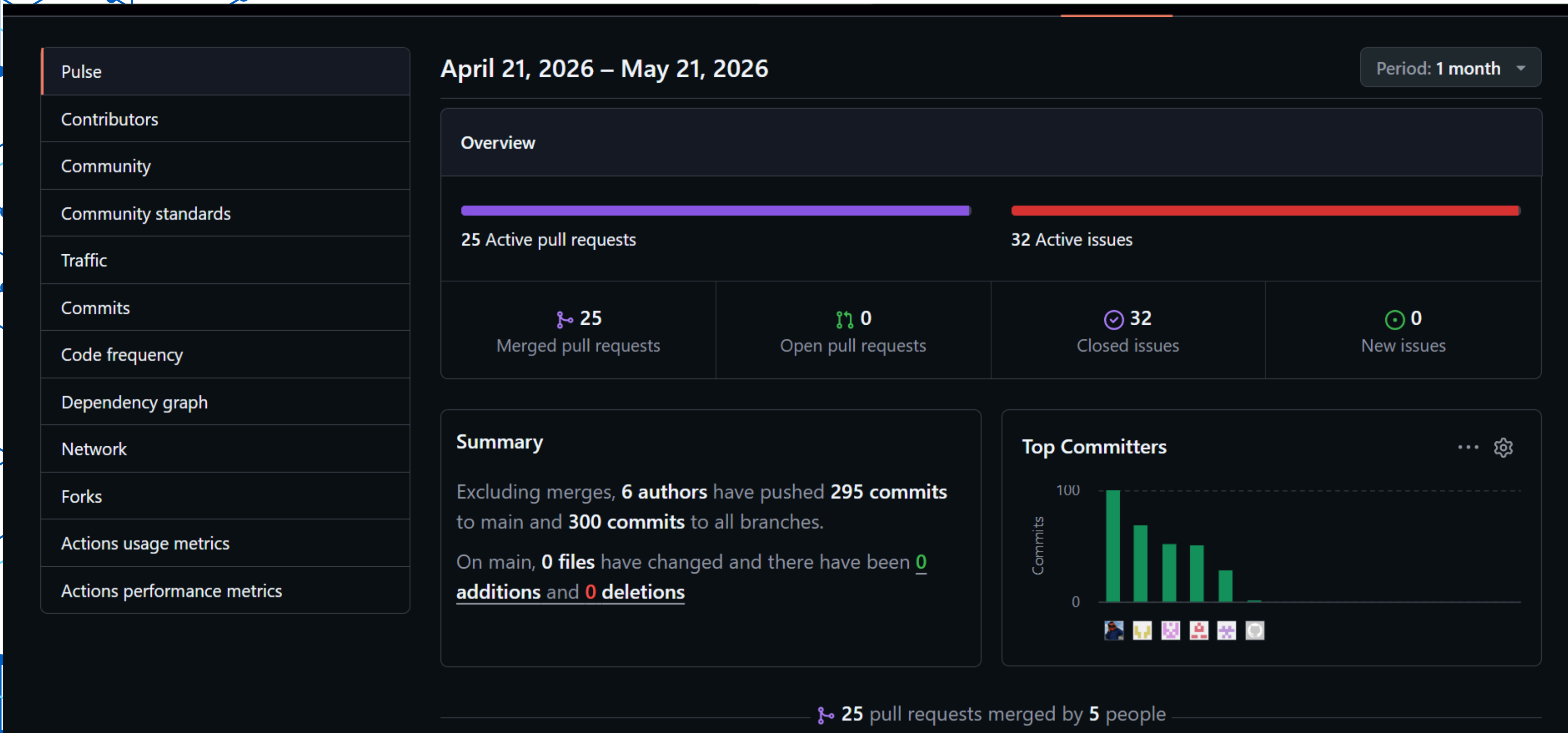
```
Ran all test suites matching src/listings/listings.service.spec.ts.
```

# GITHUB STRATEGY



- We follow **GitHub Flow**
- All the code in **main** branch will be ready for production.
- Code will be refined on the **develop** branch to ensure that it will be production ready to be pushed into the main branch.
- Feature branches will be branched from the develop branch.
- Feature branches are and will be based off the develop branch.
- Feature branches are given descriptive name of feature they're based on.
- All code is reviewed and tested before merging.
- Going forward, critical issues and changes will be committed to the develop branch.

# GROUP CONSISTENCY





# GROUP CONSISTENCY

COS301-SE-2026 / Projects / NexusDev - Uni Textbook Marketplace

Q Type / to search

NexusDev - Uni Textbook Marketplace

Add status update

Insights

Workflows 7

View 1

View 2

+ New view

Q Filter by keyword or by field

Backlog 4

This item hasn't been started

Uni-Textbook-Marketplace #26

[Frontend] Wire login page to Gift's auth API

Uni-Textbook-Marketplace #31

[Database] Write seed data for development

Uni-Textbook-Marketplace #49

Auth testing

Uni-Textbook-Marketplace #50

Listing Testing

In Progress 6

This is actively being worked on

Uni-Textbook-Marketplace #36

[Documentation] Swagger Docs + Frontend Auth Support

0 / 1 0%

Uni-Textbook-Marketplace #28

[Frontend] Help Omphemetse complete any unfinished UI work

Uni-Textbook-Marketplace #27

[Documentation] Complete SRS sections (Functional Requirements, API Contracts,

Blocked 0

Waiting on something external

In Review 4

PR open, waiting for

Uni-Textbook-Marketplace

[FEAT] Build base UI

Uni-Textbook-Marketplace

[FEAT] Build registration verification pages

Uni-Textbook-Marketplace

[FEAT] Implement a register and OTP endpoint

# WHAT'S NEXT



Privacy-first in-app messaging (Firebase Firestore microservice)



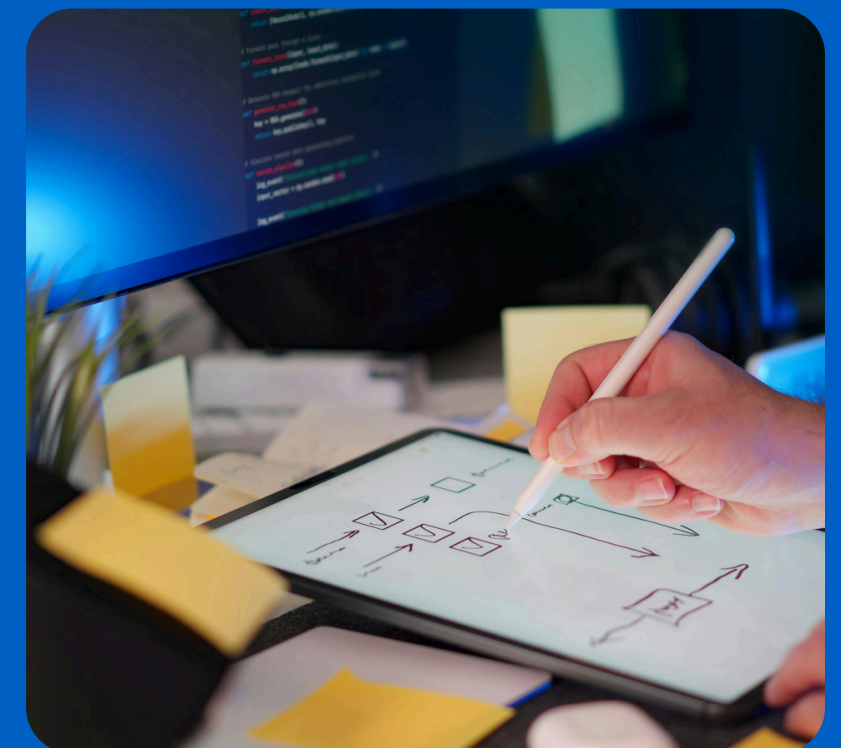
Smart search filters (price cap, edition, condition, annotation level)



Meetup ratings (successful, no-show, cancelled)



Wanted board per module





---

# THANK YOU

